NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 09-04-26**
Time: 3 Hrs **Max. Marks: 180**

09-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-4(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

TIME: 3HRS **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. z_1, z_2, z_3 are three unimodular complex numbers which are also roots of the equation $z^3 + az^2 + bz + 1 = 0$ (where a,b are complex numbers). If tangents are drawn to $|z| = 1$ at points z_1, z_2, z_3 which intersect pair wise at points $\omega_1, \omega_2, \omega_3$, then $\omega_1, \omega_2, \omega_3$ are roots of the equation

A) $(ab-1)z^3 + 2(b^2 - a)z^2 + 8bz + 8 = 0$ B) $(ab-1)z^3 + 2(b^2 + a)z^2 - 8bz + 8 = 0$

C) $(ab-1)z^3 - 2(b^2 + a)z^2 + 8bz + 8 = 0$ D) $(ab-1)z^3 + 2(b^2 + a)z^2 + 8bz + 8 = 0$

2. Let $F : (1, \infty) \to R$ be the function defined by $F(x) = \int_x^{x^2} \frac{dt}{\ln t}$.

A) F is one-to-one (i.e. injective) and the range of F is $(\ln 2, \infty)$
B) F is one-to-one (i.e. injective) and the range of F is $(1, \infty)$
C) F is many-to-one and the range of F is $(-\infty, \ln 2)$
D) F is many-to-one and the range of F is $(-\infty, -1)$

3. Let $\{b_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers such that $b_0 = 1, b_n = 2 + \sqrt{b_{n-1}} - 2\sqrt{1 + \sqrt{b_{n-1}}}$. Then $\sum_{n=1}^{\infty} b_n 2^n =$

A) $2 - \ln 2$ B) $2 - \ln 4$ C) $3 - \ln 2$ D) $3 - \ln 4$

4. Let there be a plane P passing through the line $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z+2}{3}$ and parallel to $x-$axis.
Let $\vec{v}$ be a vector, parallel to the plane P, such that the angle between $\vec{v}$ and $\vec{u} = \hat{i} + \hat{j} + \hat{k}$ is the least. Also, there is a circle lying on the plane P having centre $C(0,0,-5)$ and radius $\sqrt{10}$. A line L lying on plane P is such that it is tangent to the circle at a point $Q(x, y, z)$ having the least possible value of z. then which of the following is/are true

A) $\vec{v}.(2\hat{i} - 8\hat{j} + \hat{k}) = 1$ B) Minimum value of $|\vec{u} - \vec{v}| = \sqrt{\frac{2}{5}}$

C) The line L passes through $(2,1,-8)$ D) Q is $(0,-1,-2)$

For More Join: @IITJEE_Advanced

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. If $f:[0,1]\to[0,1]$ is an onto function such that $f'(x)>0$ and $f''(x)<0$, then

A) $\frac{f(x)}{x}\ \forall x\in(0,1]$ is a decreasing function

B) $\frac{f(x)}{x}\ \forall x\in(0,1]$ is an increasing function

C) $f(x).f^{-1}(x)\le x^2$ for some $x\in(0,1)$

D) $f^{-1}(x)\le x\le f(x), x\in[0,1]$

6. If $\sum_{n=0}^{\infty}\frac{(-1)^n}{3n+1}=\frac{1}{a}\left(\ln b+\frac{c}{\sqrt{a}}\right)$, where a, b are natural numbers then

A) $a+b$ divides $(222)^{888}+(888)^{222}$

B) $a+b$ is a prime number

C) c is a irrational number

D) $[2c]=6$ (where [.] denotes greatest integer function)

7. Which of the following statements are TRUE

A) $\int_0^1\frac{\tan^{-1}x}{x\sqrt{1-x^2}}dx=\frac{\pi}{2}\ln\left(1+\sqrt{2}\right)$

B) $\int_0^1\frac{\tan^{-1}x}{x\sqrt{1-x^2}}dx=\frac{\pi}{2}\ln\left(\sqrt{2}-1\right)$

C) $\int_0^{\frac{\pi}{2}}\log(\sin x)dx=\frac{-\pi}{2}\ln 2$

D) $\int_0^1\frac{\tan^{-1}x}{x\sqrt{1-x^2}}dx=\left(\log_2\left(\sqrt{2}-1\right)\right)\int_0^{\frac{\pi}{2}}\log(\sin x)dx$

For More Join: @IITJEE_Advanced

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. An infinite geometric sequence $\{a_n\}$ has sum $\sum_{n=0}^{\infty} a_n = 3$. The maximum possible value of the sum $\sum_{n=0}^{\infty} a_{3n}$ is

9. The ellipse $\frac{x^2}{100}+\frac{y^2}{19}=1$ is printed on a 20×20 sheet of paper so that the ellipse is centred and its axes are parallel to the edges of the paper. The sheet of paper is then rolled up into a cylinder so that the two foci of the ellipse are physically touching one other. If V is the volume of this cylinder, then $V\pi =$

10. Let $x=\sum_{n=1}^{2025}\frac{2n^2-1}{n^2(n+1)^2}$. The value of $\sqrt{x}$ is $\frac{\alpha}{\beta}$ (in reduced form) then $\alpha+\beta =$

11. The integers a and b such that $a^b = 3^0\binom{40}{0}-3^1\binom{40}{2}+3^2\binom{40}{4}-.....+3^{20}\binom{40}{40}$, $b\in(10,15)$ then $|a|+|b| =$

12. Let a normal to the ellipse $x^2+4y^2=4$ at point $(2\cos\alpha,\sin\alpha)$, $\alpha\neq n\pi, n\in I$ meet it again at point $(2\cos\beta,\sin\beta), \alpha\neq\beta$. If $\sin\beta = f(\alpha)$, then the absolute value of $\frac{17}{\sqrt{2}}f\left(\frac{\pi}{4}\right)$ is

13. Two people adopt the following procedure for deciding where to go for a cup of tea: either to a hotel or to a tea shop. Each person has a coin which has a probability $p=\frac{1}{3}$ of showing heads and q of showing tails (where $p+q=1$). In each round of the procedure, both people toss their coins once. If both coins show heads, then both people go to the hotel; if both coins show tails, then both people go to the tea shop; otherwise, they continue to the next round. This process is repeated until a decision is made.

The probability that they make a decision on the 5th round is λ then $\left[\frac{256}{100\lambda}\right] = ...$ [.] denotes GIF

For More Join: @IITJEE_Advanced

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 14 and 15:

If $f(x)$ be a continuous function over $[0,1] \to R$' such that $\int_0^1 f^2(x)dx = 1$ and $L = \int_0^1 x^{2021} f(x)dx$, then

14. Maximum value of L is $\frac{1}{\sqrt{\alpha}}$ where $\alpha =$

15. If maximum value of L is achieved for $f(x) = g(x)$, then number of points of non-differentiability of $\left(\left(\frac{x-1}{2}\right)\left(\frac{x-2}{3}\right)g(x)\right)^{\frac{1}{2023}}$ $\forall x \in (0,1)$ is/are

Paragraph-II for Question Nos. 16 and 17:

Consider the equation $ae^x = x^3$

16. The number of real solutions of the equation $ae^x = x^3$, When $a \le 0$ is β and when $a = \frac{27}{e^3}$ is γ, then $\beta + \gamma =$

17. The number of real solutions of the equation $ae^x = x^3$, when $0 < a < \frac{27}{e^3}$ is

For More Join: @IITJEE_Advanced

PHYSICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

18. In the uniform electric and magnetic fields $-E\hat{j}$ and $-B\hat{j}$ a positive charged particle is launched from the origin with an initial velocity of $v_1\hat{i}+v_2\hat{j}$. If the particle returns to the zero in its subsequent journey then: (given $E,B,v_1,v_2>0$)

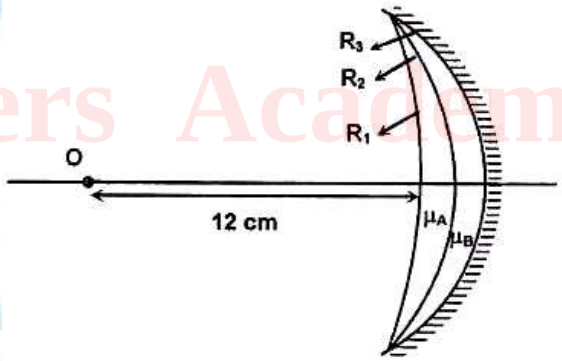
A) $\frac{v_1B}{\pi E}$ is a positive integer

B) $\frac{v_2B}{\pi E}$ is a positive integer

C) $\frac{\sqrt{v_1v_2}B}{\pi E}$ is a positive integer

D) $\frac{\left(\sqrt{v_1^2+v_2^2}\right)B}{\pi E}$ is a positive integer

19. In the adjacent figure, a thin silvered lens is shown. $\mu_A = 1.8$ and $\mu_B = 1.1$. $R_1 = 40$ cm, $R_2 = 20$ cm and $R_3 = 10$ cm. A point object is placed at a distance 12 cm from this lens on its principal axis. Then the distance of the final image formed from the lens is



A) 24cm B) 12cm C) 6cm D) 3cm

20. A particle is moving under the influence of a single force which changes with position $\vec{r}$ as,

$$\vec{F}=\frac{\left[\alpha\left|\vec{r}-\vec{b}\right|^3+\beta\left|\vec{r}-\vec{a}\right|^3\right]\vec{r}-\left[\alpha\vec{a}\left|\vec{r}-\vec{b}\right|^3+\beta\vec{b}\left|\vec{r}-\vec{a}\right|^3\right]}{\left[\left|\vec{r}-\vec{a}\right|\left|\vec{r}-\vec{b}\right|\right]^3}$$

Where α, β are positive constant and $\vec{a}$, $\vec{b}$ are constant vectors, then which of the following is true:

A) The force is non-conservative with no points of equilibria

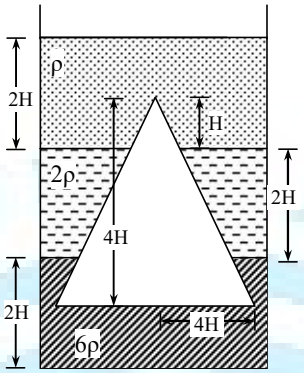
B) The force is non-conservative with a single point of equilibrium

C) The force is conservative with a single point of stable equilibrium

D) The force is conservative with a single point of unstable equilibrium

For More Join: @IITJEE_Advanced

21. A cone of height 4H and base radius 4H is in equilibrium with its axis vertical in a container containing three immiscible ideal fluids of density ρ, 2ρ and 6ρ each of height 2H as shown in the figure. The height of cone in ρ fluid is H. Neglecting any atmospheric pressure find the ratio of the force experienced by cone due to ρ fluid to the force experienced by it due to 6ρ fluid.



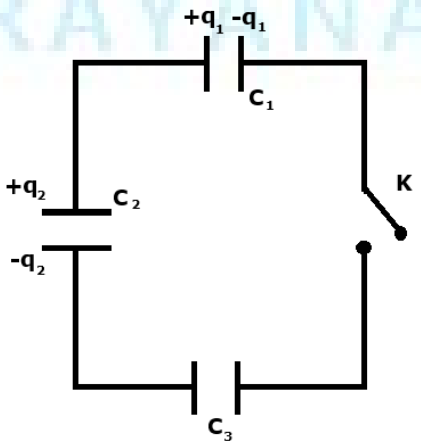
A) $\frac{1}{222}$ B) $\frac{5}{384}$ C) $\frac{1}{12}$ D) $\frac{3}{111}$

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

22. Three capacitor $C_1 = 2C$ with an initial charge $q_1 = q$, $C_2 = C$ with an initial charge $q_2 = 2q$ and a neutral $C_3 = C$ are connected as shown in the figure with an open key K. After the key is closed, which of the following statement(s) is/are incorrect after closing the key.



For More Join: @IITJEE_Advanced

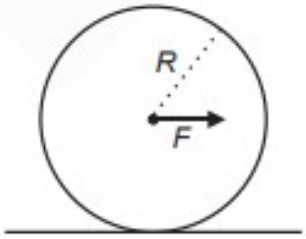
A) Value of the charge on C_1 will become $\frac{8}{5}q$

B) Heat generated in the network during the process is $\frac{9q^2}{20C}$

C) Energy lost by C_2 is $\frac{9q^2}{25C}$

D) Energy gained by C_3 is $\frac{51q^2}{50C}$

23. A horizontal force F is applied at the centre of mass of a uniform cylindrical object of mass m and radius R, perpendicular to its axis as shown in the figure. The coefficient of friction between the object and the ground is μ. The centre of mass of the object has an acceleration a. The acceleration due to gravity is g. Given that the object rolls without slipping, which of the following is/are correct?



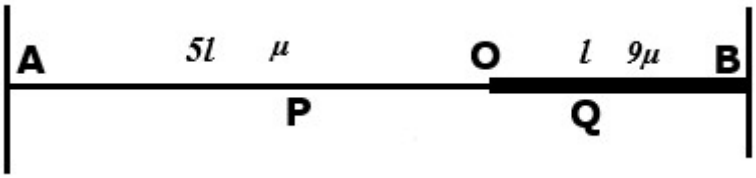
A) For the same F, the value of a does not depend on whether the cylinder is solid or hollow

B) For a solid cylinder, the maximum possible value of a is $2\mu g$

C) The magnitude of the frictional force on the object due to the ground is always μmg

D) For a thin-walled hollow cylinder, $a = \frac{F}{2m}$

24. A composite rope AOB tied to rigid supports at ends A and B is made of segments AO and OB with lengths $5l$ and l and linear mass densities μ and 9μ respectively. Standing waves are setup in this composite rope with a certain frequency such that two points P on AO segment and Q on OB segment are stationary for all time as shown in the figure. Then which of the following is/are correct:



For More Join: @IITJEE_Advanced

A) There exists integers α, β, n such that OP $= \frac{l}{5}\left(\frac{\alpha\pi+\beta}{n}\right)$, if O is stationary for all time.

B) There exists integers α, β, n such that OP $= \frac{l}{3}\left(\frac{3\alpha+\beta}{n}\right)$, if O is stationary for all time.

C) There exists integers α, β, n such that OP $= \frac{l}{3}\left(\frac{6\alpha+2\beta+4}{2n+1}\right)$, if O oscillates with highest possible amplitude.

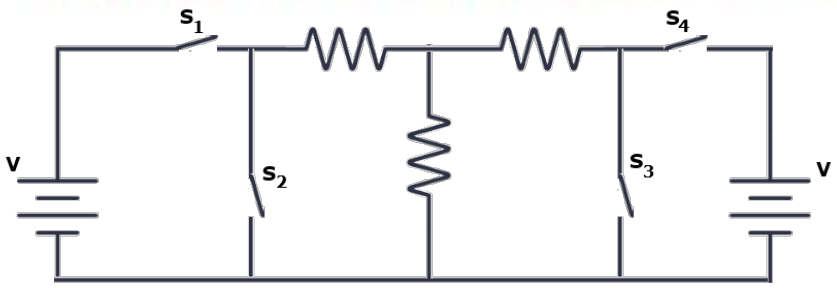
D) There exists integers α, β, n such that OP $= \frac{l}{5}\left(\frac{6\alpha+\pi\beta-1}{2n+1}\right)$, if O oscillates with highest possible amplitude.

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

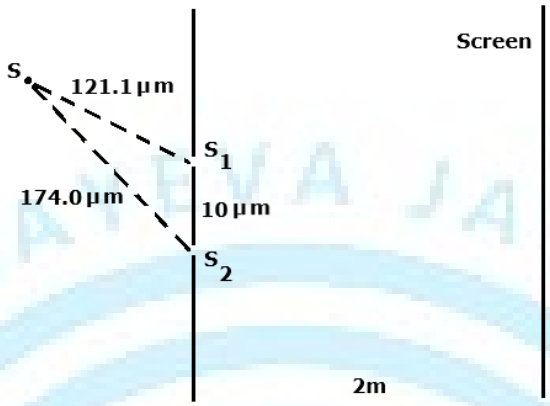
25. Consider a Vernier callipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier callipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then find the least count of the screw gauge (in microns) if the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers.

26. In the circuit shown both are identical ideal batteries. When S_1 only is closed the power liberated in the network is 1W, when S_1 and S_3 are closed together power liberated is 2W and when S_4 alone is closed the power liberated is 2W. Find the power liberated (in W) when only S_2 and S_4 are closed.



For More Join: @IITJEE_Advanced

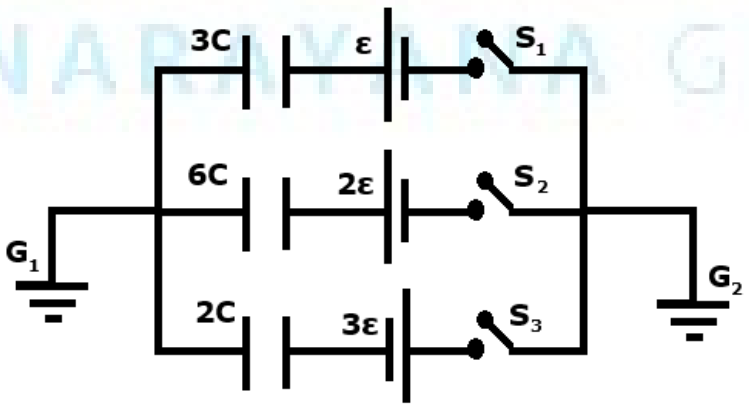
27. A monochromatic source of light S is emitting light of wave length $\lambda = 500nm$ in a YDSE setup. The distance of the source from the two slits is 121.1 μm and 174.0 μm respectively. The separation between the slits is 10 μm. If a large screen is place 2m away from the plane of the slits as shown in the figure, find the number of maxima observed on the screen.



28. An X-ray tube with a copper target is found to be emitting wavelengths other than those due to copper because of an impurity in the target. The K_α lines of the copper and the impurity have wavelengths 1.5404 A° and 0.3851 A° respectively. Find the atomic number of the impurity (Atomic number of copper is 29)

29. The force exerted by a uniform magnetic field B perpendicular to a wire having a length L and current I on the wire is given as F = ILB. An experiment plot shows F as a function of L. The plot is a straight line with a slope S = (25 ± 1) X 10^{-5} T-A and the current in the wire is (50 ± 1) mA. Find the percentage error in evaluation of B.

30. Three uncharged capacitors with open keys are connected to two grounding point G_1 and G_2 as shown in the network below. If all three switches are closed simultaneously a charge of $n\varepsilon C$ flows into the G_1 ground. Find the value of $|n|$.



For More Join: @IITJEE_Advanced

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 31 and 32:

From the surface of a barren atmosphere-less planet of mass M and radius R a small object of mass m (<<M) is launched from a point P in a non-radial direction. After a time Δt the object returns back to planet and strikes a point Q on the surface with a velocity anti-parallel to launch velocity such that the chord PQ subtends an angle 2θ at the center of the planet. If the object moves purely due to the gravitational influence of the planet, answer the following questions.

31. Given $\cos\theta = \frac{\pi}{25}$, find the value of $\frac{\Delta t\sqrt{GM}}{2\pi R^{3/2}}$

32. If K_{min} is the minimum kinetic energy attained by the small object during the journey from P to Q, then find the value of $\frac{GMm}{2RK_{min}}$, given $\cos\theta = \frac{1}{9}$

Paragraph-II for Question Nos. 33 and 34:

Two unknown devices D_1 and D_2 have the power factors of 0.6 and 0.8 respectively when connected separately across the same AC sinusoidal source of frequency f. To determine their impedances, we use a known inductor of inductance L.

33. When D_1 and L are connected in series and connected across the same source the power factor the combination of the two was found to be 0.8. If Z_1 is the minimum possible magnitude of impedance of D_1, then find the value of $\frac{2\pi fL}{Z_1}$

34. When D_2 and L are connected in series and connected across the same source the power factor the combination of the two was found to be 0.6. If Z_2 is the maximum possible magnitude of impedance of D_2, then find the value of $\frac{7Z_2}{15\pi fL}$

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

35. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult's law with a maxima in P_T vs X graph. The normal boiling point (T_b) of pure acetone is less than that of Pure CS_2. Pick out the correct statement among the following.

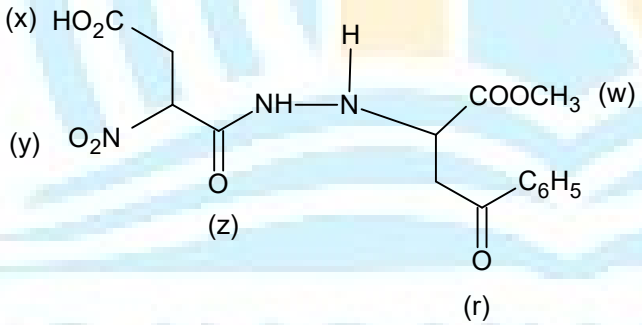
A) Boiling temperature of mixture is always less than boiling temperature of acetone.

B) Boiling temperature of azeotropic mixture is always less than boiling temperature of pure CS_2.

C) When a small amount CS_2 is added to excess of acetone boiling point of resulting mixture increases.

D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation

36. Identify the incorrect statement(s) given for the reduction of the following compounds:



A) $LiAlH_4$ reduces x, y, z, w, r

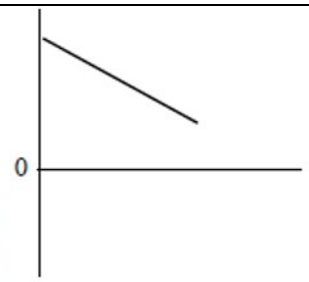
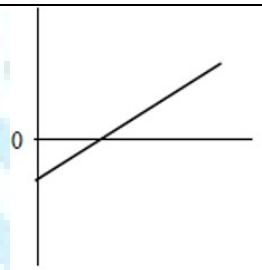
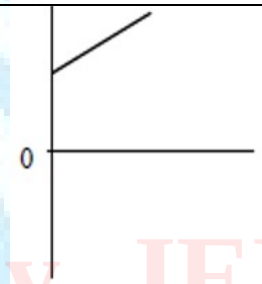
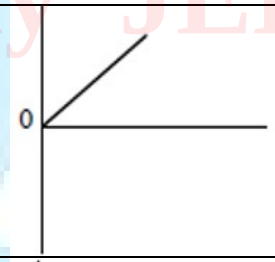
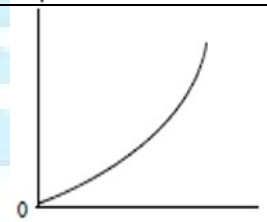
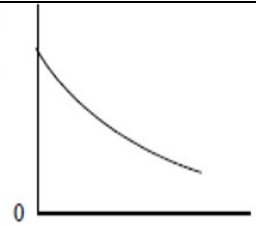
B) $NaBH_4$ reduces 'r' only

C) Na/C_2H_5OH reduces x, r and w only

D) $(CH_3)_2CHOH + \left[(CH_3)_2CHO\right]_3 Al, \Delta\, reduces only, 'r'$

For More Join: @IITJEE_Advanced

37. Match the possible outcomes that are given as qualitative plots in Set-II that represents dependence of a variable with other in Set-I

Set-I		Set-II	
1)	X-axis : $\frac{1}{T}\left(Kelvin^{-1}\right)$ Y-axis: log K K= Equilibrium constant of chemical reaction whose ΔH^0 & ΔS^0 are constant in the given temperature (T)	P)	
2)	X-axis : $1n\left[Cu^{+2}\right]$ Y-axis: E $E = E_{Cu^{+2}/Cu}$ Temperature is constant $E^0_{Cu^{+2}/Cu} = +0.34V$	Q)	
3)	X-axis : $[HCl] mol/L$ Y-axis: $\Lambda_m\left(molar\ conductivity\right)$ in Scm^2mol^{-1} Follows Debye-Huckel theory	R)	
4)	X-axis : Temperature (T) in K Y-axis: Rate Constant	S)	
		T)	
		U)	

A) 1-P, 2-R, 3- P, 4-T

B) 1-P, 2-R, 3- U, 4-T

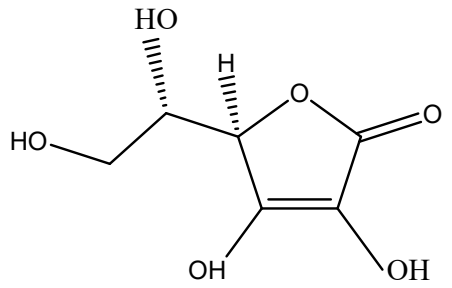
C) 1-S, 2-R, 3- P, 4-R

D) 1-P, 2-R, 3- U, 4-S

For More Join: @IITJEE_Advanced

38.

X=



the compound X is a pale yellow crystalline powder with sharp acidic taste (At wt : $C=12, O=16, H=1$) then Correct statements is/are

A) It can't liberate CO_2 gas from $NaHCO_3$

B) its deficiency lead to bleeding gums

C) the most acidic Hydrogen of OH group present on the Carbon connected to $C=O$ group

D) % of C by mass is 34

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. A solution containing Ni, Cu, and Zn prepared by dissolving in nitric acid, was reacted in separate beakers with $NaOH(aq)$ and $NH_3(aq)$, and KCN respectively. The correct statement is/ are

A) When Excess of $NaOH$ is added forms Precipitate Contain Copper and Nickel

B) When Excess of NH_3 is added form Precipitate it contain Zn ion till pH is 9.5

C) When Excess KCN is added , all are found in solution form with an oxidation of metal is +2 in their compounds

D) With excess of Ammonia is added to the solution at higher $\mathbf{P^H}$ (greater than 9.5), all metal ions in the solution form complexes with coordination number four

For More Join: @IITJEE_Advanced

40. Which of the following process (es) are correctly depicted?

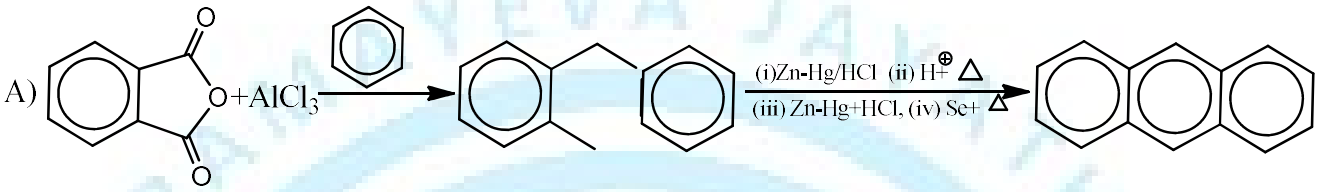
A) $ZnS \xrightarrow[\text{Floatation}]{\text{Froth}} \xrightarrow{\text{Roasting}} \xrightarrow[\text{Reduction}]{\text{Self}} \xrightarrow{\text{Distillation}} \xrightarrow[\text{Refining}]{\text{Electro}} Zn$

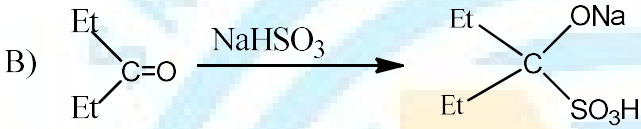
B) $SnO_2 \xrightarrow[\text{Separation}]{\text{Magnetic}} \xrightarrow[\text{Reduction}]{\text{Carbon}} \xrightarrow{\text{Poling}} \xrightarrow[\text{Reduction}]{\text{Electro}} Sn$

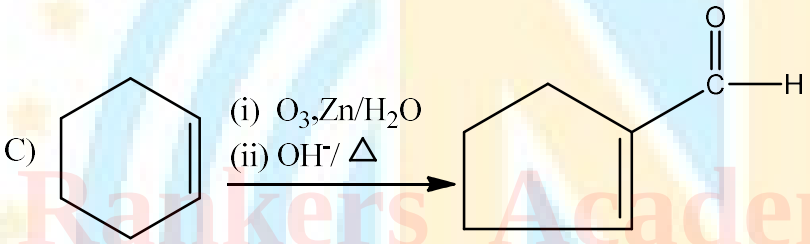
C) $Fe_2O_3 \xrightarrow[\text{Separation}]{\text{Magnetic}} \xrightarrow[\text{Reduction}]{\text{Carbon}} \xrightarrow[\text{Phase refining}]{\text{Vapour}} \xrightarrow[\text{Refining}]{\text{Electro}} Fe$

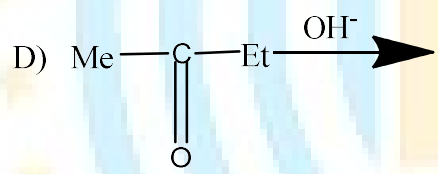
D) $TiO_2 \xrightarrow[\text{Separation}]{\text{Gravity}} \xrightarrow[\text{Reduction}]{\text{Thermite}} \xrightarrow{\text{Van-Arkel process}} Ti$

41. Which of the following reactions(s) given correct about product as shown

A) 

B) 

C) 

D) 

Total aldol reaction products including stereoisomers = 6

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

42. Consider Brass is a binary alloy with 30% Cu by mass. Varieties of Brass have wide importance because of their mechanical, chemical, electrical, and acoustic properties. Change in these properties and colours can be achieved by variation in the composition of the two metals. Brass with low zinc percentages is used for musical instruments. Higher concentration of Zn atoms (being larger in size) reduces the mobility of Cu atoms, making the alloy harder. A sample of Brass was dissolved using conc. H_2SO_4 (with catalytic amount of conc. HNO_3) to prepare a sample solution 'X', In the Iodo metric titration of solution '**X**' gives a white Precipitate. The Spin only magnetic moment of white Precipitate is **P** and Number of electrons present in d- subshell of the metal ion other than copper in the solution X is Q then find the value of $(P+Q)$

For More Join: @IITJEE_Advanced

43. $AgBr(s)$ crystallise in rock-salt type structure. At temperature T, a sample of $AgBr$ contains both frenkel and schottky defects. $2\%Ag^+$ ions of ideal crystal exhibit frenkel defect and due to frenkel defect all interstitial Ag^+ ions occupy tetrahedral voids. 0.5% of formula unit $(AgCl)$ are completely missing the crystal lattice due to schottky defect .The crystal remains electrically neutral.
Density of perfect $AgBr$ crystal is ρ_0 and density of imperfect crystal is ρ.
If perfect crystal contains N formula units of $AgBr$, then how many of the following information's are correct?

i) Total number of Ag^+ ions at normal sites are 0.975N.
ii) Total number of Ag^+ at interstitial sites are 0.02N.
iii)Total number of Br^- Ions present are 0.995N.
iv) Value of $\frac{\rho}{\rho_0} = 0.975$
v) Fraction of tetrahedral voids occupies by $Ag^+ = 0.01$ (ignore loss of tetrahedral voids due to defects)
vi) Coordination number of interstitial Ag^+ ions due to Frenkel defects is 4

44. The overall formation constant of $\left[Co(NH_3)_6\right]^{2+}$ in aqueous solution is 10^5 and the standard potential for reduction of $Co^{3+}(aq)$ and $\left[Co(NH_3)_6\right]^{3+}(aq)$ are as follows

$Co^{3+}(aq) + e \rightleftharpoons Co^{2+}(aq) \qquad E^o = 1.90V$

$\left[Co(NH_3)_6\right]^{3+}(aq) + e \rightleftharpoons \left[Co(NH_3)_6\right]^{2+}_{(aq)} \quad E^o = +0.1V$

If the nearest overall formation constant of the $\left[Co(NH_3)_6\right]^{3+}$ ion is 10^X. Then the value of **x** would be (Given : $E_{cell} = E^o_{cell} - 0.06\log Q$)

45. The product obtained at anode by the electrolytic oxidation of 50% H_2SO_4 solution at high current density is treated with Aq. Manganese (II) solution gives a manganese containing product (P) .Number of moles of faintly alkaline solution of 'P' required to oxidise completely 15 moles of thio sulphate is

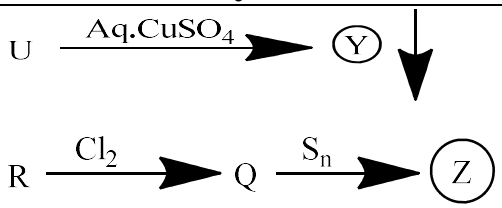
46. $P_4 + SOCl_2 \rightarrow R + \underset{(Oxide)}{S} + T$

'T' is used in vulcanisation of rubber, R is Phosphorous containing binary compound

$T + Cl_2 \rightarrow W \xrightarrow{C_2H_4} \underset{Explosive}{X}$

$P_4 + NaOH \rightarrow \underset{(\text{Re}duced\ product)}{U} + \underset{(Oxidised\ product)}{V}$

For More Join: @IITJEE_Advanced



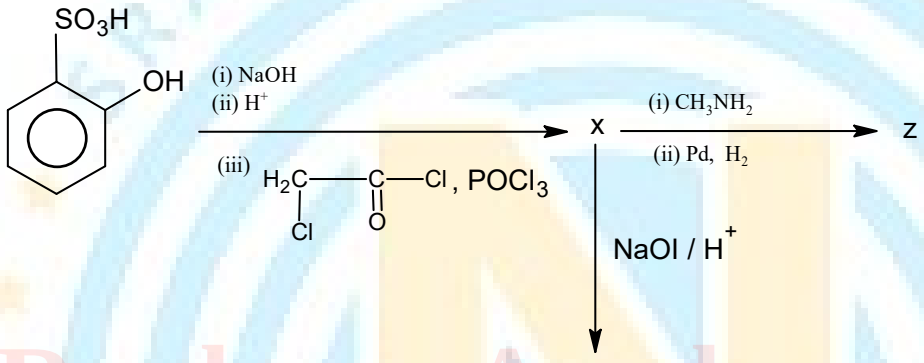
$A =$ Oxidation state of metal in Z

$B =$ Total number of Sp^2 hybridised atoms in 'X'

$C =$ Quantity of mass contributed by the non metallic part in 1 mole of **y** (in gram)

Find the value of $(\mathbf{A}+\mathbf{B}+\mathbf{C})$, (Atomic weight: $Cu = 64, P = 31, S = 32, Cl = 35.5, C = 12, H = 1, O = 16$)

47.

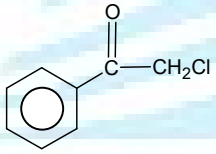
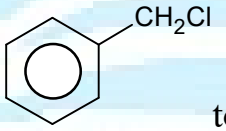


Total number of correct statements is /are

(i) The compound '**y**' is having four oxygen atoms per molecule

(ii) The compound '**z**' contain two chiral atoms per molecule

(iii) The compound '**z**' has ketone group, 2^0 amine and phenolic –OH groups

(iv)  is more reactive than  towards Nucleophilic substitution reactions towards given nucleophile

(v) The compound '**y**' gives positive response with $NaHCO_3$ and also with neutral $FeCl_3$

(vi) The compound '**z**' reacts with active metals and also gives positive response with ceric ammonium nitrate

(vii) The compound '**y**' is more acidic than Resorcinol

For More Join: @IITJEE_Advanced

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 48 and 49:

Two equilibria are simultaneously existing in a vessel at 25^0 C

$NO(g)+NO_2(g) \rightleftharpoons N_2O_3(g)\ K_{P_1}$

$2NO_2(g) \rightleftharpoons N_2O_4(g)\ K_{P_2}=8atm^{-1}$

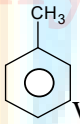
Initially only $NO(g)$ and $NO_2(g)$ are present in 3:5 mole ratio. The total pressure at equilibrium is 5.5 atm and the partial pressure of NO_2 at equilibrium is 0.5 atm.

48. K_{P_1} for the equilibrium (atm^{-1}) is

49. Partial pressure of NO at equilibrium (atm) is

Paragraph-II for Question Nos. 50 and 51:

Scheme-1:

The compound 'P' is obtained by vigorous Oxidation of 

with alkaline $KMnO_4$ followed by acidification

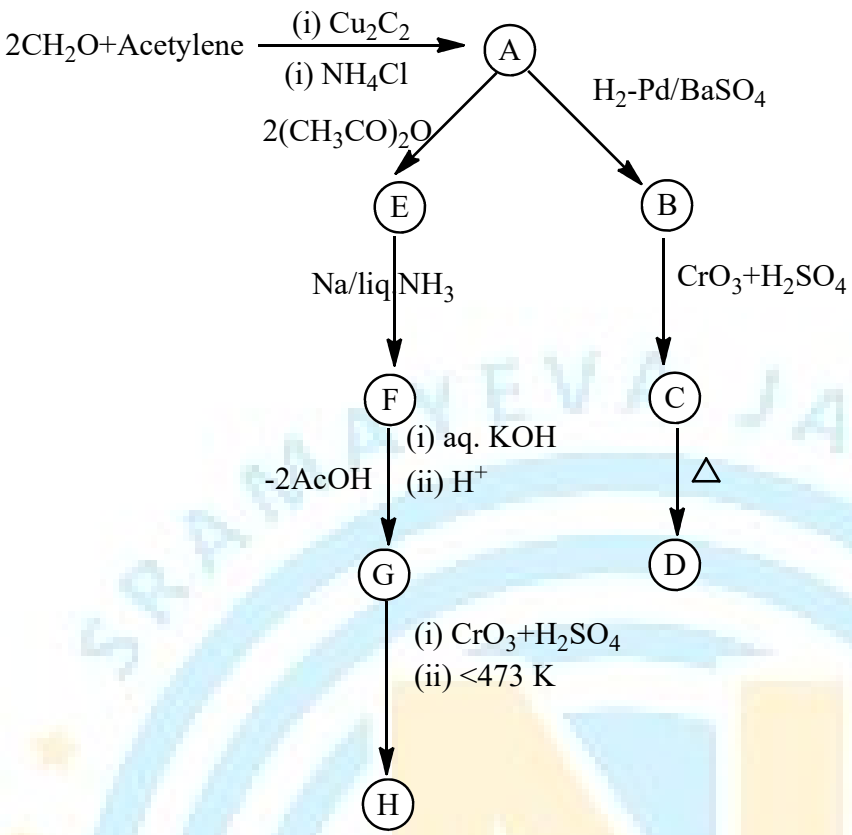
$$P \xrightarrow[\substack{(ii)P_4O_{10}\\(iii)Na-Hg/C_2H_5OH}]{(i)NH_3,\Delta} X \xrightarrow[(ii)CrO_3+H_2SO_4]{(i)HNO_2} P$$

Scheme-2:

$$P \xrightarrow[\substack{(ii)NH_3\\(iii)NaOH+Br_2}]{(i)SOCl_2} Y \xrightarrow[\substack{(ii)HNO_2+HCl,278K\\(iii)NaOH}]{(i)Br_{2(Excess)}/H_2O} Q$$

For More Join: @IITJEE_Advanced

Scheme -3:



50. Number of correct statements in the following about X,Y,P,Q

a) '**X**' is more basic than **Y**

b) '**X**' is more reactive than **Y** towards EAS

c) **X** and **Y** can be differentiated by using Hinesburg's reagent

d) The compound 'Q' reacts with Benzene diazonium salt form a coloured dye

e) Total mass of product (g) obtained when one mole of Q reacts with one mole of P is 435

f) Gabriel Pthalimide synthesis is used to prepare X and Y

51. **Answer the question based on Scheme-3. Find the Value of** $(\mathbf{R}+\mathbf{S})$

Total number of Oxygen atoms Present in D and H is R

Total number of Pi bonds B and E is S

For More Join: @IITJEE_Advanced